wheel and axle gear seats are the same as those on freight and passenger locomotive axles.

Because of possible dangers resulting from axle cracks, it is recommended that axles be given a Magnaglo or Magnaflux test whenever a wheel is removed, regardless of length of service. The roller bearing inner races also should be removed so the ends of the axle under the bearing race can be visually inspected for fretting and Magnaflux tested. (See "Magnaflux Inspection" at end of this instruction.)

Crack indications on an axle in the transverse plane are dangerous and the axle should be scrapped, unless individual railroad specifications permit reworking to entirely remove the crack. The axle should be scrapped if suspension bearing journals or the journals of axles for friction type boxes show evidence of bearing seizure accompanied by thermal cracks. Copper penetration makes the axle steel very brittle and likely to break. Under no condition should this axle be reused until inspected for copper penetration and approved for use by the Railroad Engineer of Tests.

Axles with runout up to a maximum of .030" total indicator reading with journals at zero, may be continued in service with no detrimental effect on suspension or journal bearings provided that a thorough Magnaflux inspection of the axle has shown that no defects exist. If the axle has over .030" total runout, it should be scrapped.

Axle dimensions, new and condemning limits, are shown in Fig. 4.

Axles and axle gears may be used until worn to condemning limits, if Magnaflux inspection reveals no defects.

If axles are not Magnafluxed at each wheel removal, it is recommended that the axle be scrapped after two pair of wheels have been worn to their condemning limit.

In order to insure accurate axle measurements, the micrometers used should be checked against a known standard. Axles should always be measured when they are cold, never when they are warm from machining. All fillets and axle ends must be polished free from tool marks, and each time wheels are turned, all burrs or other irregularities should be polished off axle ends. If it is necessary to machine axle ends, the same amount should be removed from each end to assure that the axle is centered in the truck after assembly.

If the axle has a spline for a speed governor drive, the spline should be visually inspected and replaced if any of the teeth are found broken, chipped, cracked or with rounded edges.

The journals should also be checked for taper and remachined if the taper exceeds 1/32" for the length of the journal. The microfinish must be maintained.

If the wheel set has been removed from a truck involved in a wreck, the inner races must be removed and the journals of the axles should be Magnaglo or Magnaflux tested.

Axle Journal - Roller Bearing

The minimum and maximum axle journal diameter is to be governed by the fit between the axle journal and the inner race. IT IS IMPERATIVE that the interference fit be not less than .003". This is to insure that the inner race will not work loose on the axle or be overstressed.

The absolute minimum diameter of an axle journal should not be less than 6.502", and the I.D. of the race applied to this axle should not be greater than 6.499".

When the .003" to .005" interference fit cannot be obtained on a used axle, the journal can be machined to dimensions shown in Fig. 5, for use with a .020" undersized race.